Hint and answer:

Case(a)

Pulley is fixed rigidly here. When the mass displace by y and the spring will also stretch by y. Therefore, $F = T = ky$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Case(b)

Mass displace by y, pulley also displaces by y. $T = 4ky$.

$$T = 2\pi\sqrt{\frac{m}{4k}}$$

BOOKS FOR REFERENCE

1. Vibrations and Waves – A. P. French, CBS publisher and Distributors Pvt. Ltd.
2. Concepts of Physics – H. C. Verma, Volume 1 and Volume 2, Bharati Bhawan Publisher.
3. Fundamentals of Physics – Halliday, Resnick and Walker, Wiley Publishers, 10th edition.
4. Physics for Scientist and Engineers with Modern Physics – Serway and Jewett, Brook/ Coole Publishers, Eighth Edition.

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